



MATERIAL TEST CERTIFICATE

EN 10204 Type 3.1 — Mill Test Report

CERTIFICATE NO.

MTC-AAM-2026-0847

Purchase Order: PO-APD-2026-004517

CoC Ref: CoC-AAM-2026-0451

EN 10204 — 3.1

MANUFACTURER Apex Aerospace Materials Ltd.	CUSTOMER AeroPrecision Dynamics Inc.	CERTIFICATE DATE 2026-08-13
MILL / ORIGIN Sheffield Titanium Works Ltd.	MILL CERT REF MC-STW-26-8821	DELIVERY REFERENCE PACK-LIST-2026-00742

MATERIAL 1 OF 3

APD-TF-7721-R6 — Turbine Disc Forging — Ti-6Al-4V (AMS 4928)

Heat: HT-2026-TI-4418

MATERIAL TRACEABILITY

Material origin: Sheffield Titanium Works Ltd. — Melt: VIM/VAR (Double Vacuum) | Ingot: ING-TI-26-4418 | Forged at: Apex Aerospace, Derby

ELEMENT	AL	V	FE	O	C	N	H	Y	TI
AMS 4928 Min	5.50	3.50	—	—	—	—	—	—	Rem
AMS 4928 Max	6.75	4.50	0.30	0.20	0.08	0.05	0.015	0.005	—
Heat Result	6.12	4.01	0.18	0.16	0.02	0.01	0.008	<0.001	Bal
Status	✓	✓	✓	✓	✓	✓	✓	✓	✓
PROPERTY	UTS (MPA)	YIELD 0.2% (MPA)		ELONGATION (%)		RA (%)	HARDNESS (HRC)		
AMS 4928 Min	895	825		10		20	—		
Test Result	948	879		14.2		36.8	33		
Status	✓ PASS	✓ PASS		✓ PASS		✓ PASS	✓ PASS		

MATERIAL 2 OF 3

APD-IB-3318-K2 — Inconel 718 Bar Stock (AMS 5663)

Heat: HT-2026-IN-7791

MATERIAL TRACEABILITY

Material origin: Special Metals Corp. (Hereford, UK) — Melt: VIM/VAR | Ingot: ING-IN-26-7791 | Heat Treated per AMS 2774 at Apex Aerospace

ELEMENT	NI	CR	NB	MO	TI	AL	CO	FE
AMS 5663 Min	50.0	17.0	4.75	2.80	0.65	0.20	—	Rem
AMS 5663 Max	55.0	21.0	5.50	3.30	1.15	0.80	1.00	—
Heat Result	52.8	18.6	5.12	3.04	0.92	0.54	0.31	Bal
Status	✓	✓	✓	✓	✓	✓	✓	✓
PROPERTY	UTS (MPA)	YIELD 0.2% (MPA)		ELONGATION (%)		RA (%)	HARDNESS (HRC)	
AMS 5663 Min	1240	1035		12		15	—	
Test Result	1312	1089		16.5		22.1	40	
Status	✓ PASS	✓ PASS		✓ PASS		✓ PASS	✓ PASS	

APD-SH-0094-M1 — Mounting Bracket — 7075-T6 Al Alloy (AMS 4045)

Lot: LT-2026-AL-2290

MATERIAL TRACEABILITY

Material origin: Constellium — Isabel, VA, USA — Lot: LT-2026-AL-2290 | Heat treated to T6 temper per AMS 2770 at Apex Aerospace

ELEMENT	ZN	MG	CU	CR	FE	SI	MN	AL
AMS 4045 Min	5.1	2.1	1.2	0.18	—	—	—	Rem
AMS 4045 Max	6.1	2.9	2.0	0.28	0.50	0.40	0.30	—
Lot Result	5.64	2.48	1.61	0.22	0.18	0.12	0.04	Bal
Status	✓	✓	✓	✓	✓	✓	✓	✓
PROPERTY	UTS (MPA)		YIELD 0.2% (MPA)		ELONGATION (%)		HARDNESS (HB)	
AMS 4045 Min	524		455		7		—	
Test Result	562		493		10.8		150	
Status	✓ PASS		✓ PASS		✓ PASS		✓ PASS	

MATERIAL TEST LAB — AUTHORIZED SIGNATORY

INDEPENDENT INSPECTION AUTHORITY

Dr. Sarah Holden

Head of Quality & Metallurgy
Date: 2026-08-13

TÜV Rheinland — Inspector #4417

Third-Party Verification
Date: 2026-08-13